BEFORE THE NEW YORK STATE
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	**Case 19-G-0309**

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations Of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	**Case 19-G-0310**

DIRECT TESTIMONY OF KIM FRACZEK AND LEE ZIESCHE ON BEHALF OF SANE ENERGY PROJECT

August 30, 2019

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I. INTRODUCTIONS AND BACKGROUND

Q.Please state your name and business address.

A. Kim Fraczek and Lee Ziesche. 232 E 11th Street, New York, NY 10003

Q. What is your occupation?

A:(FRACZEK) Director of Sane Energy Project
(ZIESCHE) Community Engagement Coordinator of Sane Energy Project

Q.Have you previously testified before the Commission?

A:Yes. We recently provided direct testimony for the Con Edison rate case. We have also provided written comments and attended multiple PSC public hearings on numerous PSC cases and have helped New Yorkers understand the public commenting processes and submit thousands of comments on numerous cases before the PSC on energy issues.

Q. What is Sane Energy Project

A. Sane Energy Project is a non-profit organization representing 8,500 New Yorkers. We are based in New York City, and work with communities across New York State to stop the buildout of fracked gas infrastructure and hasten the just transition to renewable energy. Since 2011, we have worked directly with scientists, health professionals, and impacted communities across the state to raise awareness about the health and climate impacts of fracked gas infrastructure, and we work with communities to identify and remove barriers to community-owned renewable energy. We've worked with thousands of New Yorkers to prepare and provide public comment and participate in the regulatory process for over a dozen fracked gas infrastructure projects across New York.

Q.Please summarize your background.

A:(FRACZEK) I have been an employee of Sane Energy Project since 2014. Prior to this, was a volunteer coordinator for Sane Energy Project since its inception in 2011. My background is 12 years in corporate creative direction and digital rapid prototyping leading teams of designers in New York, USA, Tokyo, Japan, Hong Kong and Visakhapatnam, India in the fashion industry, to produce products for the American Market. I began volunteering my time to remedy water quality issues with people in Pennsylvania, where my family lives, and built a non-profit organization to focus on this work full-time. I have traveled extensively in Pennsylvania and

New York to organize with communities, scientists, medical doctors and elected officials to change laws that are lenient toward the fossil fuel industry that compromises human health, wildlife health and air and water quality detriment.

(ZIESCHE) I have a bachelors in Telecommunications from the University of Florida School of Journalism with a minor in International Development and Humanitarian Assistance. I am currently in post-production for my full length documentary directorial debut. For three and a half years I was employed as the Grassroots Coordinator for International WOW Company's award winning GASLAND films. In that time I traveled around the world to communities being impacted by fracking and fracked gas infrastructure with a focus on communities in Pennsylvania and New York and often organized events with leading experts on the health and climate impacts of shale gas development. Since 2017 I've been working as the Community Engagement Coordinator for Sane Energy Project. During that time I've continued to travel to communities in New York and Pennsylvania being impacted by fracked gas infrastructure and produced over a dozen short documentary films telling their stories.

Q: What is your purpose in submitting this testimony?

A: The purpose of the testimony is to address the failure of the Company to present a proposal that takes into account the unique climate and health impacts of fracked gas and the growing public demand to get off fossil fuels. The Company is proposing to spend millions of ratepayer dollars on Storm Hardening while at the same time investing hundreds of millions of dollars on fossil fuel infrastructure that will only make future storms stronger by contributing to climate change.

II. GAS ISSUES

A. Introduction and Background

Q: What will you address relating to the Company's business activities?

A. We have reviewed the Company's policy overview of its gas business activities, and this testimony addresses four issues relating to these activities.

First, we oppose the Company's proposals to spend hundreds of millions of ratepayer dollars on gas capital investments to replace Leak Prone Pipeline (LPP) instead of retire it and convert customers from oil to gas because these proposals conflict with state and city climate goals when the unique climate impacts of fracked gas are taken into account.

Second, we oppose the Company's proposal to invest hundreds of million of ratepayer dollars to extend the life of LNG and CNG facilities.

Third, we'd like to address that National Grid's dependence on the Williams NESE Pipeline shows a deep disregard for public demand around energy and incorrect calculations of fracked gas greenhouse emissions.

Fourth, taking into account the unique health impacts of shale gas we request funds be used for radon testing to insure the safety of the Company's service.

Q. What is the overarching context in which the Company's gas business activities must be evaluated?

A.Shale gas production in the Appalachia region has increased rapidly since 2012 meaning an increasing amount of the gas supply coming into New York is extracted from the Marcellus Shale using high volume hydraulic fracking. Nationally, by 2016, 69%[1] of all wells drilled in this country were fracked and that number has continued to rise. The proposed Williams NESE Pipeline that much of the Company's rate case proposal supports would be transporting fracked gas from Pennsylvania. And in her Direct Testimony[2] Elizabeth D. Arangio confirmed more of the Company's gas will be fracked gas saying "as supplies from the Marcellus shale region became abundant and readily accessible, the Companies did not renew expiring long-haul contracts with Union, TransCanada, and Empire pipelines that delivered more expensive supplies from Dawn, Canada." Because of the extreme method used to develop gas from the Marcellus Shale, fracked gas has greater environmental impacts for the climate and on human health[3] than conventional gas and must be evaluated differently.

B. Gas Proposals Conflict with City & State Greenhouse Gas Reduction Goals

Q.What kind of gas capital expenses is the Company proposing?

A.The company is proposing to replace and expand Leak Prone Pipeline (LPP) and other pipeline through its Proactive Main Replacement Program, Gas System Reinforcement Program

[1] See EIA *Hydraulically fractured horizontal wells account for most new oil and natural gas wells* https://www.eia.gov/todayinenergy/detail.php?id=33972

[2] Direct Testimony of Elizabeth D. Arangio before the Public Service Commission, April 2019 (p. 20)

[3] The *Compendium of Scientific, Medical, and Media Findings Demonstrating Risks and Harms of Fracking (*the Compendium*)* is a fully referenced compilation of the evidence outlining the risks and harms of fracking. https://concernedhealthny.org/compendium/

and Metropolitan Reliability Infrastructure (MRI) Program. In its Future of Heat Panel testimony the company says it will replace 225 miles per year for a total of 930 miles over four years. In its Future of Heat Panel testimony the Company is also proposing to increase gas demand by converting oil customers to fracked gas through its Customer Connections Programs adding 7,000 new customers. The Company claims both these programs will help them meet New York State greenhouse gas reduction goals and their own methane reduction goals of 60% by 2035.

Q.What are New York State greenhouse gas reduction goals?

A:The New York State Energy plan sets a 40% greenhouse gas reduction target for 2030 and 80% reduction by 2050. The recently passed and signed into law Climate Leadership and Community Protection Act (CLCPA) also requires 40 percent emissions reduction by 2030 and an increased requirement of 85 percent emissions reductions by 2050.

Q:What are the city's goals greenhouse gas emission reduction goals?

A: Intro 1253 passed by the New York City Council requires buildings over 25,000 square feet to reduce greenhouse gas emissions by 40% by 2030 and 80% by 2050.

Q: Can the Company expand gas capacity and meet state and city greenhouse gas emissions reduction goals?

A.No. When the full life cycle of methane leakage is taken into account, gas expansion goes against city and state goals to reduce greenhouse gas emissions. Methane, the main component of gas is a powerful greenhouse gas that is 86-106 times more potent a warming agent than C02 for the first 20 years it's in the atmosphere. FLIR footage has shown high leakage rates of methane at the wellhead and all along the transportation route at compressor and metering stations. Cornell scientist Robert Howarth estimates that when looking at upstream and downstream emissions from shale gas, leakage rates range between 3.8% and 12%[4]. Even if upstream emissions were not taken into account Howarth estimates the downstream emissions associated with storing gas and delivering shale gas to market to be 2.5%. With leakage rates like this it is impossible to meet city and state goals to reduce greenhouse gas emissions and is more likely taking us in the wrong direction by increasing emissions.

[4] See Robert W Howarth *Methane emissions and climatic warming risk from hydraulic fracturing and shale gas development: implications for policy* https://www.eeb.cornell.edu/howarth/publications/f_EECT-61539-perspectives-on-air-emissions-of-methane-and-climatic-warmin_100815_27470.pdf

Q.How are the Company's assertions about meeting greenhouse gas emission goals inadequate?

A. When it comes to oil to gas conversions through the Customer Connections Program, the company primarily bases its claims of reducing greenhouse gas emissions on the fact that when burned, gas emits less CO2 than oil, but ignores upstream and downstream methane leakage. In doing so they are presenting an incomplete and misleading representation of greenhouse gas emissions associated with their projects and their environmental impact. Additional when talking about reducing greenhouse gas emissions the Company proposes projects to reduce methane leaks[5] within their system while at the same time expanding gas capacity and extending the overall life use of pipelines through replacements. Expanding and extending gas capacity will lead to more fracking and fracking infrastructure needed to transport the gas, thus increasing overall methane emissions.

Q: Does the Company's project proposals to limit methane address the issue?

A: No. While identifying and fixing leaks is important, the technologies proposed to identify leaks do not address the upstream emissions especially when paired with pipeline and infrastructure enhancements that will overall increase gas use.

Q:What should the Company do instead of investing in gas capital project to replace pipeline and convert customers?

A:The Company should not invest any new ratepayer funds in expanding gas capacity and extending the life of pipelines and should instead invest in gas demand reduction and beneficial electrification. The company should look at retiring LPP instead of replacing it and convert customers to non-gas alternatives.

C. LNG & CNG Reliability

Q:What is the company proposing?

A: The company is proposing the following projects as investments to create LNG and CNG reliability: Compressed Natural Gas Special Projects, LNG Blanket, LNG – Special Projects,LNG TANK 1 and Tank 2 Upgrade Projects, LNG - Vaporizers 7 & 8 Replacement and LNG - Salt Water Pump House.

[5] Future of Heat Panel

Q:Do you support this investment? Why or why not?

A: No. As stated in above testimony the increased dependency and expansion of fracked gas goes against New York policy to reduce greenhouse gas emissions. We need to plan for shutting these facilities down rather than investing hundreds of millions of dollars on infrastructure that will increase our reliance on dirty fossil fuels.

Q: In addition to climate impacts are there other reasons the Company should plan on closing LNG & CNG facilities instead of investing in them?

A: Dangerous to surrounding communities and workers. LNG is highly volatile and explosive, and notably New York City saw a disaster that killed 40 workers[6].

Q: What do you recommend?

A: The Company should not invest any new funds that would extend the use of LNG and CNG facilities and should instead invest in gas demand reduction.

D.Williams NESE Pipeline

Q:Does the Company propose projects connected to the approval of the Williams Northeast Supply Enhancement Pipeline?

A: Yes. According to the Company the proposed Marine Park Regulator Station Project is required to "utilize the increased capacity from the Transco Northeast Supply Enhancement project."[7] The Company also claims in its testimony that the pipeline is critical to meet increase demand in its service territory and it would suspend its customer connections programs if the pipeline is not approved[8].

Q:Do you support investments connected to the Williams Northeast Supply Enhancement Pipeline? Why or why not?

A: No. As mentioned in above testimony, climate science and city and state law require a dramatic reduction of greenhouse gases and it's impossible to lower greenhouse gas emissions and expand fracked gas infrastructure at the same time. Additionally New Yorkers, including many of the Company's ratepayers, have overwhelming called on New York State to reject

[6] https://www.silive.com/news/2013/02/40_years_ago_today_staten_isla.html
[7] Exhibit ___ (GIOP-5) Page 90 of 120
[8] Gas Infrastructure and Operations Panel - Pages 118 & 119

Williams Transco's application to build the Northeast Supply Enhancement (NESE) project. In their temporary denial of the project in May the New York Department of Environmental Conversation said 14,000 public comments were submitted on the project and over 90% were opposed to the pipeline. Since then New Yorkers have submitted an additional 25,000 public comments opposing NESE on the pipeline's reapplication and a recent poll[9] found the more New Yorkers learn about the pipeline, the more likely they are to oppose, and in fact by a margin of nearly 5-to-1, New Yorkers would rather have the state expand clean energy solutions over dirty fossil fuels. The Company's continued push for NESE project goes against what New Yorkers ratepayers are demanding and against city and state climate goals. Additionally, according to a study[10] done by PSE Health Energy for Earthworks the NESE project would cause an estimated 7.8 million metric tons of CO2 equivalent per year (in their mid-range modelling scenario). New York City as a whole produces about 50 million metric tons of CO2 equivalent. This pipeline alone would produce so much climate pollution that it would be equivalent to roughly 15% of New York City's greenhouse gas emissions taken as a whole or roughly about 4% of the entire State of New York's greenhouse gas emissions. The Company's claims that the NESE project and related investments in gas expansion would lower greenhouse gas emissions and meet state and city goals is simply false.

D: Need for Radon Testing

Q: What about the Company's proposal that fracked gas expansion has "environmental benefits" is inaccurate?

A: The Company's claim their Customer Gas Connections provides "environmental benefits"[11] and reduces local emission fails to address the unique health threats that come with gas fracked from the Marcellus Shale. The Marcellus Shale in northeastern Pennsylvania has higher than average levels of uranium and thorium. These elements break down to form radium-226. Signs of radon in gas & other fracked gas chemicals in our homes, have been on the rise since 2004[12] when high-volume hydraulic fractured drilling began. Radon is the leading cause of lung cancer in non-smokers in the United States[13].

Q What is your recommendation?

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[9] https://www.nrdc.org/experts/rich-schrader/new-poll-new-yorkers-reject-williams-pipeline
[10] https://www.psehealthyenergy.org/wp-content/uploads/2018/02/NY.Pipelines-1.pdf
[11] Exhibit ___ (GIOP-5) Page 1 of 120
[12] Increased Levels of Radon in Pennsylvania Homes Correspond to Onset of Fracking Johns Hopkins Bloomberg School of Public Health
https://www.jhsph.edu/news/news-releases/2015/increased-levels-of-radon-in-pennsylvania-homes-correspond-to-onset-of-fracking.html
[13] https://www.cancer.org/cancer/cancer-causes/radiation-exposure/radon.html

In order for the Company to comply with its requirement to provide safe service we recommend the Company provide EPA scientific radon monitoring[14] in all gas supplied to customers. Additionally, we recommend meeting with New York elected officials who drafted the 2014 Radon Bill[15] to monitor and mitigate radon in our gas supply.

III. PROCEDURAL RECOMMENDATION: PUBLIC PARTICIPATION

Q:What barriers have you experienced in participating in this proceeding?

A: As two part-time employees working for an organization, it has been very difficult for Sane Energy Project to have the capacity to draft our own testimony and financially impossible for us to hire our own experts. Additionally the public we work with has had no opportunity to bring evidence to the table or ask questions of the Company.

Q:What is your procedural recommendation?

A: We recommend Intervenor Funds available for the public and since it is our rate payer dollars National Grid is proposing to spend, the PSC should require utilities to fund the public to hire our own experts. We also recommend multiple public hearings in locations where energy burdens are greatest earlier in the process so members of the public can provide evidence before settlement negotiations begin. We also recommend discovery be available to all members of the public not just individuals who have the capacity to intervene as a party. The climate crisis and unwanted expansion of fracked gas infrastructure have made New Yorkers more aware of and wanting to shape the energy system of New York than ever before.

[14] https://www.epa.gov/radon
[15] https://nyassembly.gov/leg/?default_fld=&bn=A06863&term=2013&Summary=Y&Text=Y